

Smart Energy Analytics – Hourly Forecasting and Weather-Vulnerability Profiling

DALLAS Project

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Abstract

We develop a reproducible pipeline to forecast next-hour electricity consumption and quantify temperature-driven vulnerability at the building level. The pipeline builds a weather-aligned hourly dataset, creates leakage-safe time features, trains and validates tree-based models with time-aware splits, and computes temperature elasticity to segment buildings by weather sensitivity. The final artifacts include trained models, metrics, explainability plots, and a vulnerability report. This report focuses on practical pipeline design and evaluation rather than novel modeling techniques.

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1 Introduction

This project develops a forecasting pipeline for next-hour electricity consumption using hourly usage and weather data from multiple regions. We engineer leakage-safe temporal and exogenous features, train tree-based models with chronological splits, and analyze predictions at both global and per-building levels. Temperature sensitivity is quantified using elasticity, enabling comparison of weather-driven responses across buildings.

Normalization Rationale Electricity usage is normalized on a per-building basis to remove scale effects and enable meaningful comparison across heterogeneous buildings. This ensures that model performance and elasticity reflect relative building behavior rather than absolute load magnitude.

2 Pipeline Overview

Our pipeline spans data discovery to deployment to reflect how forecasting systems are built and operated in practice:

1. Source discovery and acquisition (APIs, public archives, compliant scraping) with schema validation.
2. Unification and weather integration: align usage to hourly cadence; inner-join hourly weather; derive holidays.
3. Leakage-safe feature engineering: per-building lag_1h, lag_24h, rollmean_24h; calendar and weather drivers.
4. Time-aware splitting: chronological train/validation/test that mirror deployment.
5. Model training and evaluation: strong baselines and boosted trees with diagnostics.
6. Interpretability and vulnerability profiling: permutation importance, ICE/PDP, temperature elasticity and segments.
7. Packaging and reporting: reproducible scripts, plots, and programmatic report generation.
8. Testing and deployment: CLI for batch prediction, monitoring considerations for drift and stability.

3 Problem Definition and Scope

3.1 Problem Definition

We predict next-hour normalized electricity usage at the building level and quantify each building's sensitivity to temperature. Accurate short-horizon forecasts support operational tasks such as load balancing and peak management, while temperature elasticity provides an interpretable measure to identify buildings more exposed to weather extremes. The analysis focuses on a one-hour forecasting horizon to balance operational relevance and leakage control.

3.2 Scope and Data Acquisition Strategy

The study focuses on hourly data across multiple regions. To preserve interpretability in temperature-related analyses, only hours with aligned weather observations are retained. Data sources are selected for consistent schemas and clear time semantics. Model evaluation relies on chronological splits that emulate deployment.

The scope is grounded in the following use cases:

- **Operational planning:** next-hour forecasts support load balancing and peak management.
- **Targeted interventions:** high-sensitivity buildings (elasticity) for demand response and outreach.
- **Policy evaluation:** track the distribution of elasticity across time and customer segments.

4 Data Acquisition and Dataset Characteristics

4.1 Data Source and Collection Methodology

We combined two public sources (Ausgrid NSW and London LCL) and standardized them to hourly alignment, merging per-region weather (apparent temperature, precipitation, day/night). We retained only rows with weather present to avoid exogenous imputations. Identifiers are anonymized. The unified dataset serves as the single source of truth for feature engineering and chronological evaluation.

In practice, we first surveyed public archives and APIs for smart-meter or feeder-level usage data across regions. Where APIs were unavailable, we used compliant scraping to collect candidate datasets and validated schemas for usage fields, time resolution, and coverage. Several sources were excluded due to missing usage fields or incompatible granularities. We converged on Ausgrid (NSW, AU) and London LCL as consistent sources. Weather and holiday signals were integrated per region: hourly apparent temperature, precipitation, and day/night flags (temperature normalized), with public holidays derived programmatically (e.g., AU NSW, GB calendars). We kept only rows with weather present to avoid imputing exogenous drivers, preserving integrity for temperature-sensitivity analyses. This filtering reduces sample size but avoids introducing artificial assumptions about exogenous drivers, which is particularly important when interpreting temperature sensitivity.

Item	Value
Rows	74498698
Buildings	3835
Time range	2007-01-02 00:00:00 to 2022-04-30 22:00:00
Train end	2014-01-26 15:00:00
Validation end	2017-06-17 05:00:00
Test period	> 2017-06-17 05:00:00

Table 1: Dataset scope and chronological cutoffs.

4.2 Technical Challenges and Solutions

- **Heterogeneous schemas:** Usage data from different sources (CSV vs. Parquet) were normalized into a common long format (building-hour rows) to ensure consistent downstream processing.
- **Weather alignment:** Usage records were aligned to hourly weather data using strict inner joins. Rows without weather observations were removed to preserve interpretability in temperature-related analyses.

- **Leakage control:** Lagged and rolling features were computed per building using shift operations, and all evaluations relied on chronological splits to prevent look-ahead bias.
- **Scale and performance:** Due to dataset size, chunked reads and representative sampling were used for EDA, and figures and tables were generated in compact forms to maintain clarity.

These choices ensure leakage-safe modeling and preserve interpretability under realistic data constraints.

4.3 Feature Engineering

Group	Features
Calendar	hour, day_of_week, month, season, is_weekend, is_holiday
Dynamics	lag_1h, lag_24h, rollmean_24h (leakage-safe)
Weather	apparent_temperature_norm, precipitation, is_day
Target	y_next (next-hour usage per building)

Table 2: Feature groups used in modeling.

Weather Normalization Apparent temperature is normalized to the $[0, 1]$ range within each region to stabilize model training and enable consistent interpretation of temperature effects. This scaling allows temperature elasticity to be compared across buildings without conflating absolute climate differences between regions.

5 Experimental Design and Splits

We estimate time cutoffs via reservoir sampling of timestamps to approximate 80%/10%/10% train/val/test boundaries.

Random splits were deliberately avoided, as they would introduce temporal leakage and lead to overly optimistic estimates that do not reflect deployment conditions. MAE is used as the primary metric to capture typical error, while RMSE highlights sensitivity to larger deviations.

Train end=2014-01-26 15:00:00, **Validation end**=2017-06-17 05:00:00; Test strictly later. Metrics: MAE (primary), RMSE (secondary).

6 Model Development, Training, and Evaluation

6.1 Feature Selection, Preprocessing, and Initial Analysis

We curated features that reflect operational reality and avoid leakage: per-building lag_1h and lag_24h capture persistence and daily cycle; rollmean_24h summarizes recent regime; calendar features anchor seasonality and hour-level structure; weather features (normalized apparent temperature, precipitation, is_day) act as exogenous drivers. Preprocessing is minimal by design (trees are scale-robust), focusing effort on correct time alignment and splits.

6.2 Baselines and Primary Model

- Naive: $\hat{y}(t) = y(t - 1)$

- RandomForestRegressor (stable baseline)
- HistGradientBoostingRegressor (primary)

Training uses large sampled subsets per split to bound memory while preserving time ordering.

7 Key Findings

The proposed pipeline improves next-hour forecasting accuracy relative to the naive baseline, with boosted trees reducing MAE and RMSE on both validation and test periods. Temperature and recent usage dynamics emerge as dominant predictors, and building-level elasticity reveals wide variation in temperature response across buildings.

8 Results

8.1 Model Performance Evaluation

Table 3 summarizes MAE and RMSE on the validation and test periods.

Model	MAE (val)	RMSE (val)	MAE (test)	RMSE (test)
Naive	0.0384	0.0535	0.0478	0.0625
Random Forest	0.0239	0.0332	0.0330	0.0453
HistGBR	0.0209	0.0287	0.0298	0.0410

Table 3: Overall accuracy across models (val/test). Primary model: HistGBR.

8.2 Feature Importance (Validation)

Permutation importance highlights lag_1h, hour, lag_24h, rollmean_24h, day_of_week as leading drivers. Higher values indicate stronger contribution to reducing validation MAE; the prominence of temperature and recent usage aligns with expected physical and behavioral effects.

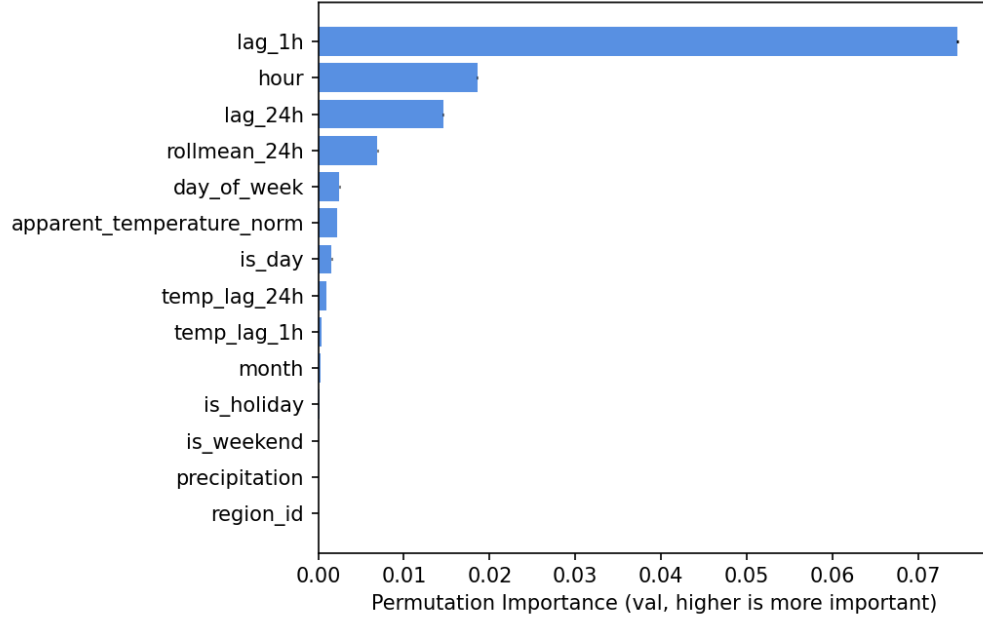


Figure 1: Permutation importance (validation) for HistGBR. Dominant features confirm the role of temperature and daily dynamics.

Top features (if computed):

Feature	Importance (mean)	Std
lag_1h	0.075	0.000
hour	0.019	0.000
lag_24h	0.015	0.000
rollmean_24h	0.007	0.000
day_of_week	0.002	0.000
apparent_temperature_norm	0.002	0.000
is_day	0.002	0.000
temp_lag_24h	0.001	0.000
temp_lag_1h	0.000	0.000
month	0.000	0.000
is_holiday	0.000	0.000
is_weekend	0.000	0.000
precipitation	0.000	0.000
region_id	0.000	0.000

Table 4: Top features by permutation importance (validation).

8.3 Fit Quality and Error Patterns

The scatter below should concentrate around the diagonal if the model is well calibrated; dispersion at higher usage levels reflects the heavier-tailed regime where short spikes are harder to capture.

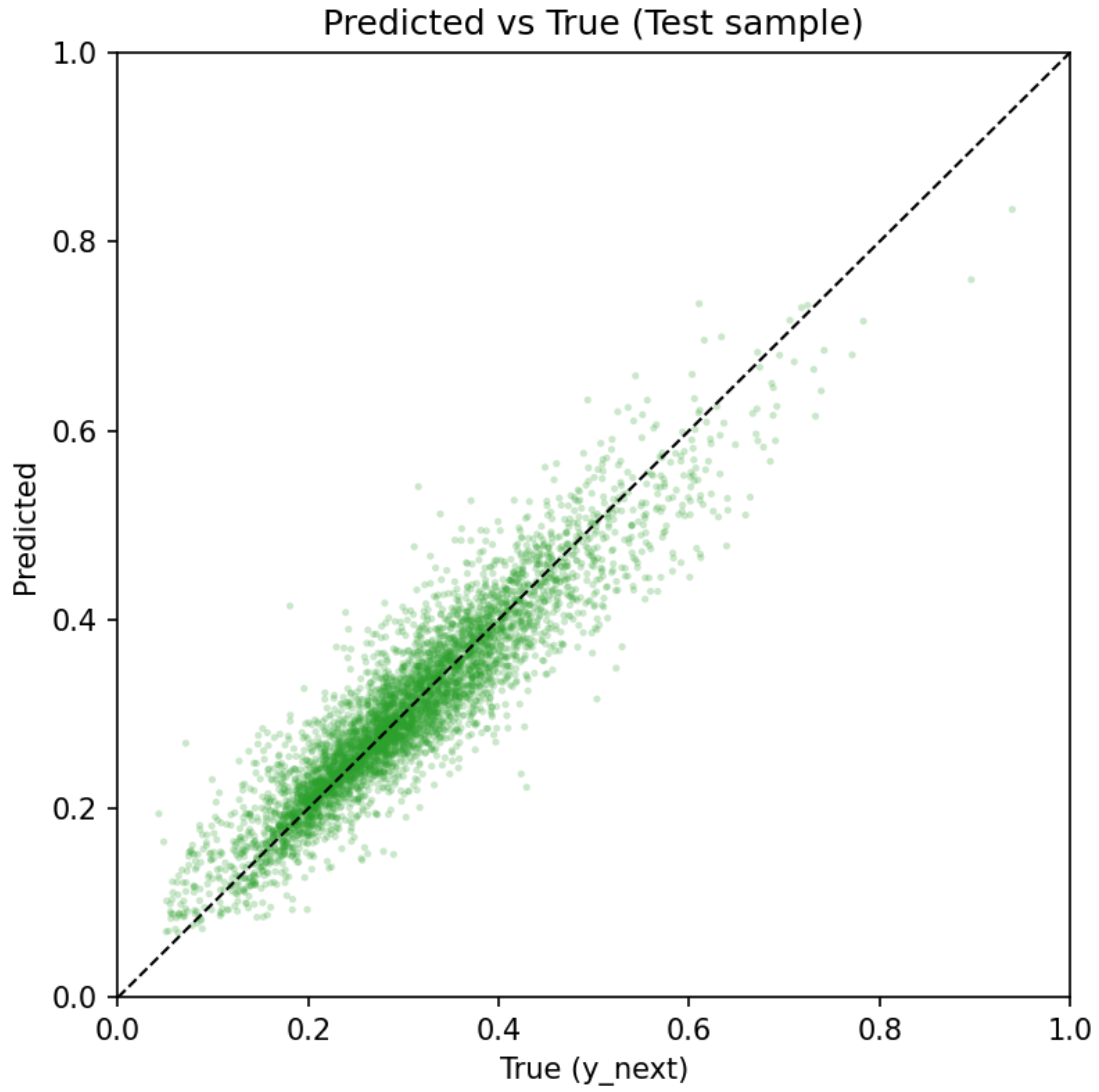


Figure 2: Predicted vs True (test sample). Points concentrate near the diagonal, while dispersion increases at higher usage levels, reflecting short-term demand spikes.

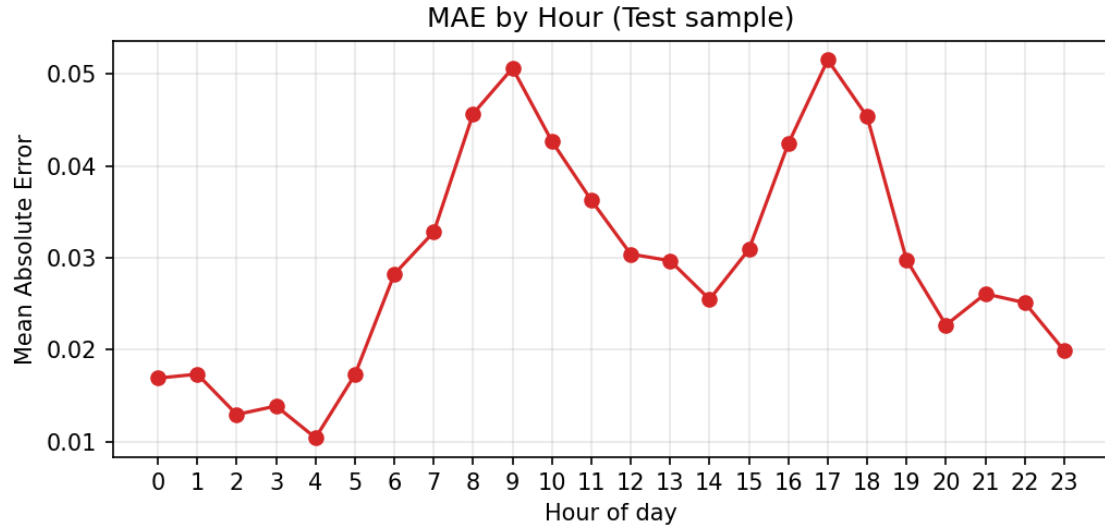


Figure 3: MAE by hour of day on the test sample. Elevated errors around transition hours (morning/evening) indicate regime changes are hardest to predict.

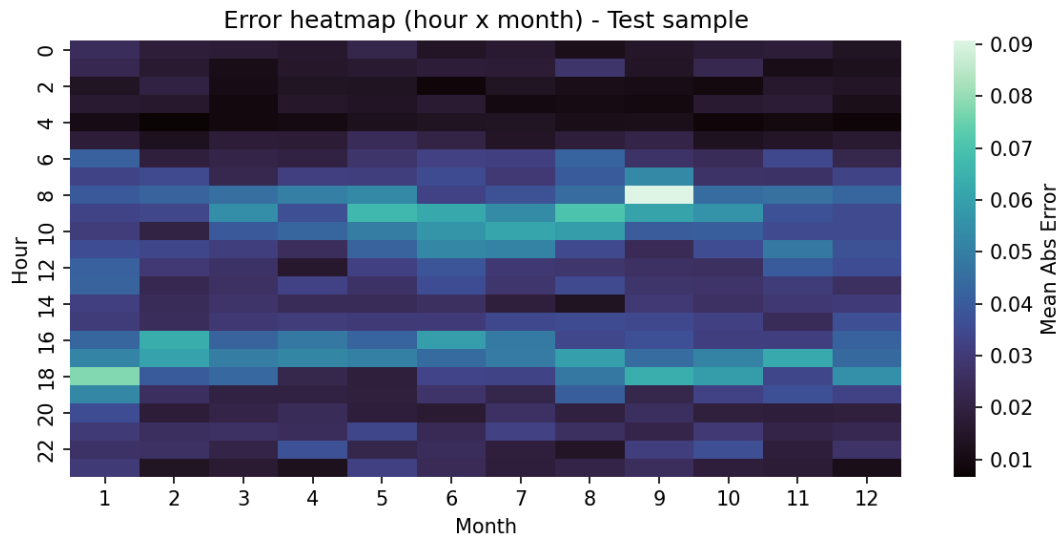


Figure 4: Error heatmap by hour and month (test sample). Warm bands reveal seasonal-hourly regimes where performance degrades, guiding targeted feature or segment refinements.

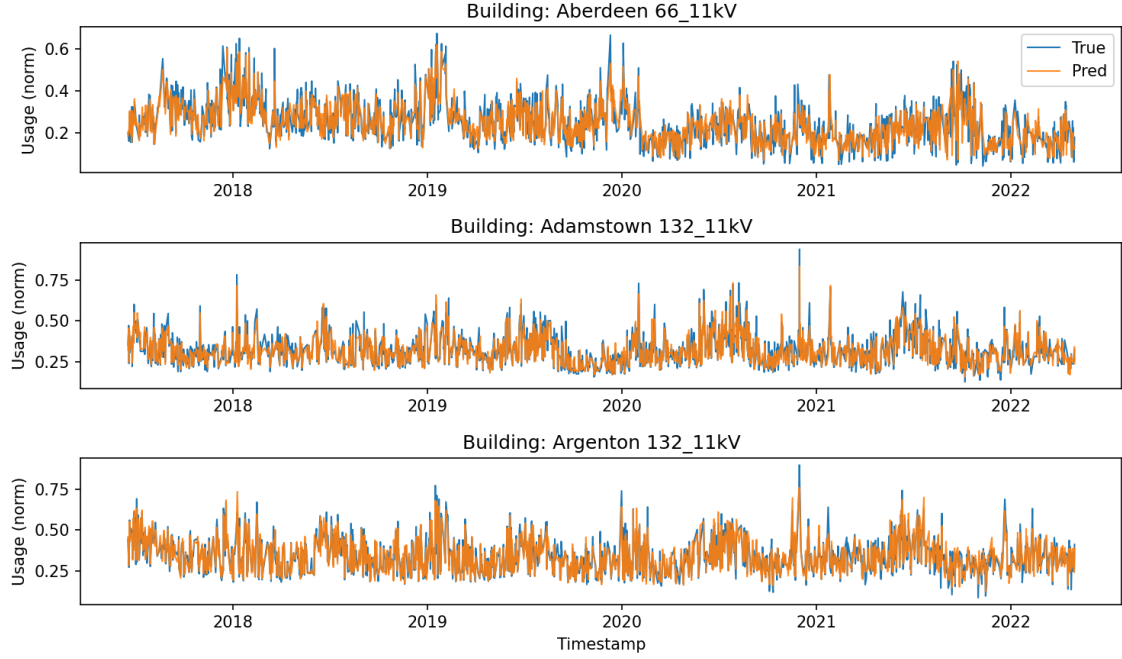


Figure 5: Example time series: predicted vs actual for top buildings in sample. The model tracks levels and turning points with small lag except during sharp spikes.

Summary of prediction behavior Across these diagnostics, predictions align with the identity line and track short-term demand dynamics under typical conditions. Remaining errors concentrate in spike-like regimes and transition periods, which are difficult to capture with short-horizon autoregressive features. This motivates the subsequent interpretability and vulnerability analysis.

8.4 Per-Building Test MAE Summary

$n=3$, $\min=0.0273$, $Q1=0.0282$, $\text{median}=0.0291$, $Q3=0.0310$, $\max=0.0330$.

9 Model Interpretability via Permutation Importance and PDP/ICE

We interpret the trained model using global feature importance and response analyses based on permutation importance and PDP/ICE. While SHAP provides additive attributions, the chosen methods offer comparable directional insights for tree-based models with substantially lower computational cost. Building-level sensitivity is summarized through temperature elasticity, capturing how predicted demand responds to changes in temperature.

9.1 Method

We estimate per-building temperature elasticity by varying `apparent_temperature_norm` on a fixed grid $[0, 1]$ for sampled hours in the test period, predicting with the trained model, and fitting a linear slope $dy/dtemp$ to the resulting ICE curve. The elasticity for each building is defined as the median slope across sampled hours, which reduces sensitivity to outliers and provides a robust summary of typical temperature response. Elasticity represents model-implied temperature sensitivity and is not interpreted as a causal effect.

9.2 Segments and Distribution

Segments by quantiles: Low=48, Moderate=94, High=48. Median elasticity=0.0108; Q1=0.0076, Q3=0.0149 (scored 190/190 buildings).

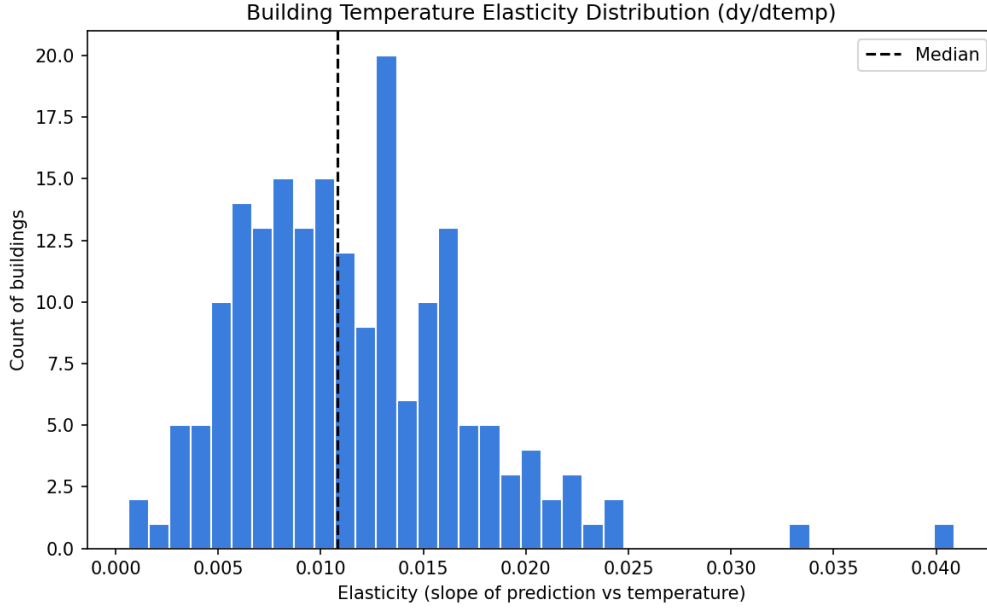


Figure 6: Distribution of building temperature elasticity (dy/dtemp).

The distribution shows a unimodal but right-skewed pattern, indicating that most buildings exhibit moderate temperature sensitivity, while a smaller subset responds much more strongly. This motivates segmenting buildings by elasticity to distinguish typical behavior from high-risk, weather-sensitive cases.

9.3 Top/Bottom Temperature Sensitivity (Test Period)

To illustrate how elasticity can be used in practice, we report representative examples from the highest and lowest sensitivity groups. These tables are not exhaustive, but highlight the range of responses observed across buildings in the test period.

Top 12 High-Sensitivity Buildings	
Building	Elasticity
Mt Hutton 33_11kV	0.0409
Port Botany 33_11kV	0.0329
Somersby 132_11kV	0.0247
Darling Harbour 132_11kV	0.0238
Paddington 33_11kV	0.0237
St Peters 132_11kV	0.0225
Revesby 132_11kV	0.0219
Berowra 132_11kV	0.0218
Hornsby 132_11kV	0.0216
Lisarow 33_11kV	0.0210
Kirrawee 132_11kV	0.0207
Kingsford 132_11kV	0.0207

Table 5: Highest positive elasticity (dy/dtemp).

Top 12 Low-Sensitivity Buildings	
Building	Elasticity
Crows Nest 33_11kV	0.0006
Tomago 33_11kV	0.0012
Tomalpin 33_11kV	0.0021
Milperra 132_11kV	0.0028
Tighes Hill 33_11kV	0.0029
Mitchells Flat 66_11kV	0.0031
Engadine 33_11kV	0.0033
Avoca 66_11kV	0.0036
Toronto 33_11kV	0.0037
Broadmeadow 33_11kV	0.0038
Carrington 33_11kV	0.0039
City East 33_11kV	0.0043

Table 6: Lowest elasticity (dy/dtemp).

10 Exploratory Data Analysis (EDA)

How to read the EDA figures The EDA diagnostics are grouped into four purposes: (i) data integrity (missingness), (ii) demand regimes (distribution and seasonality), (iii) feature relationships (correlation and partial dependence), and (iv) failure modes (error slices by hour, season, and temperature). For each group, we state the main takeaway and its implication for feature design or vulnerability interpretation.

10.1 Summary of EDA Findings

- Usage exhibits clear diurnal patterns and moderate seasonality, with smaller weekend and holiday effects.
- Errors are largest during transition hours and in hotter temperature bins in some regions.
- Engineered features show limited multicollinearity, which is suitable for tree-based models.

10.2 Extended EDA and Diagnostics (Selected Figures)

The following figures validate data quality and illustrate demand regimes, feature relationships, and error patterns relevant to next-hour forecasting.

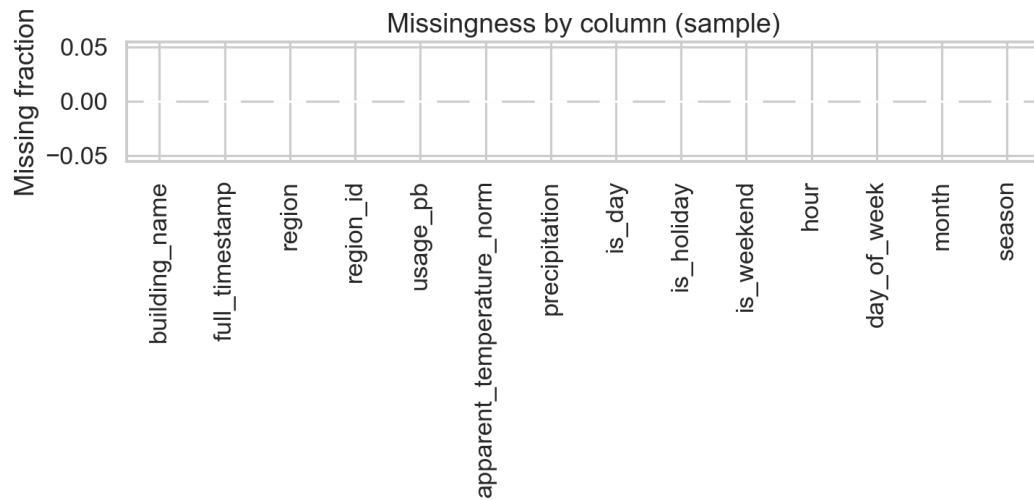


Figure 7: Missingness by column (sample).

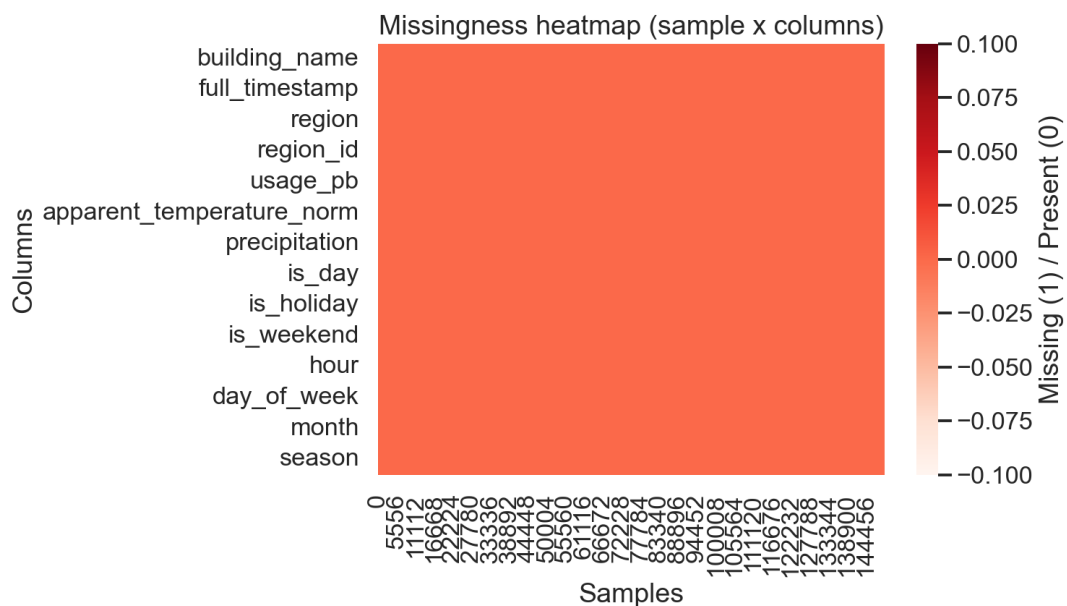


Figure 8: Missingness heatmap (sample x columns).

Takeaway (Missingness) The missingness diagnostics indicate that the modeling table is largely complete after the weather-aligned inner join, limiting the need for imputation that could bias elasticity estimates.

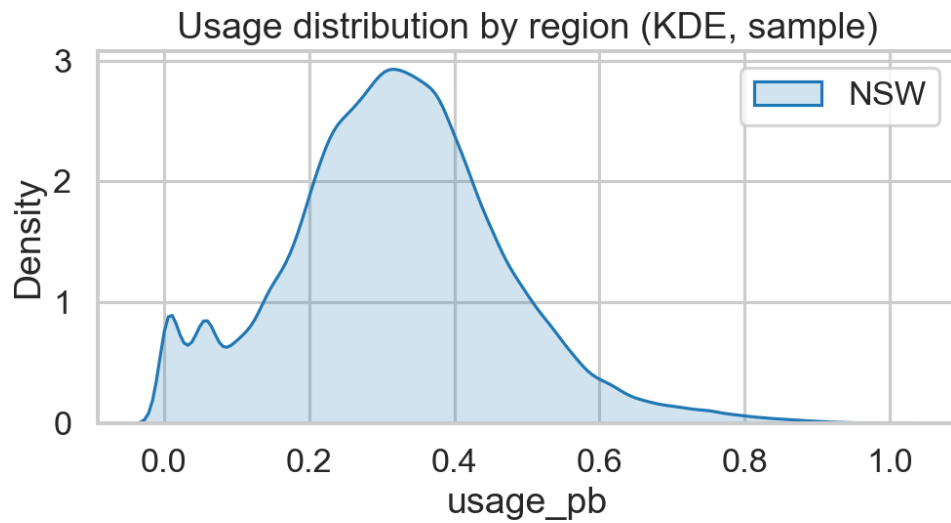


Figure 9: Usage distribution by region (KDE).

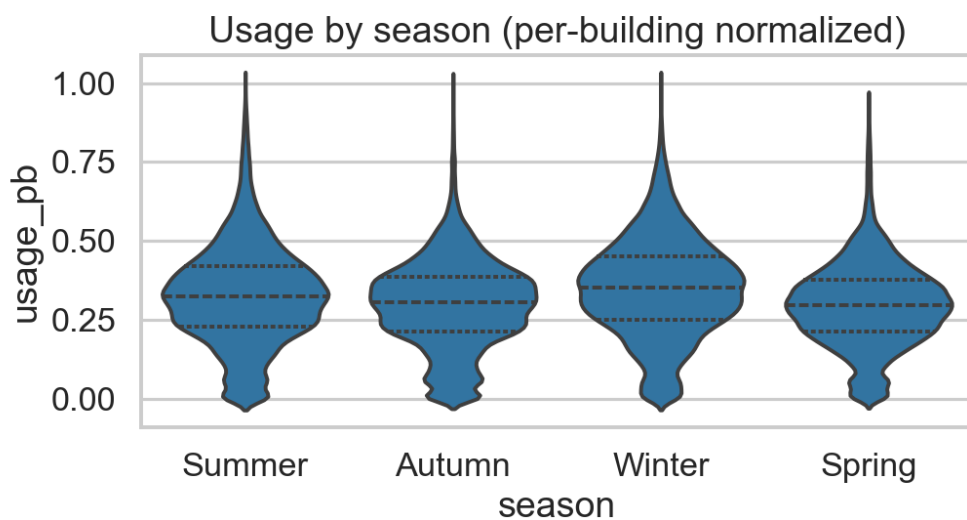


Figure 10: Usage by season (per-building normalized).

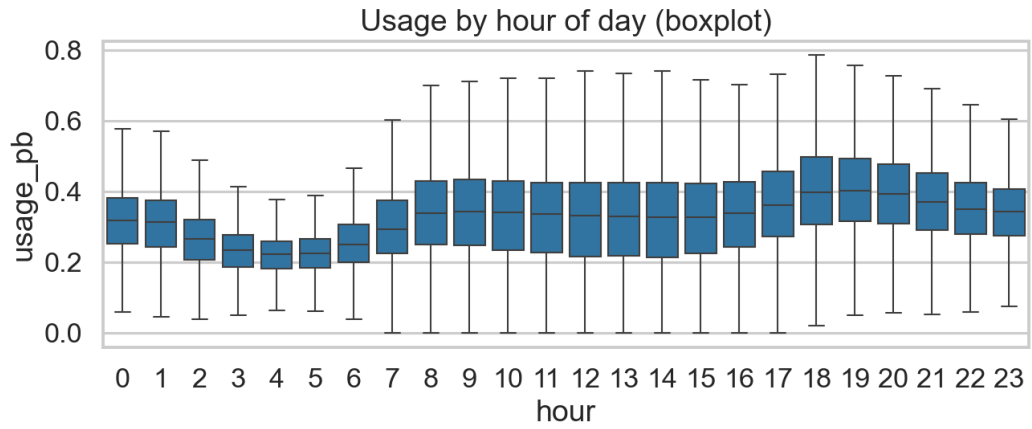


Figure 11: Usage by hour of day (boxplot).

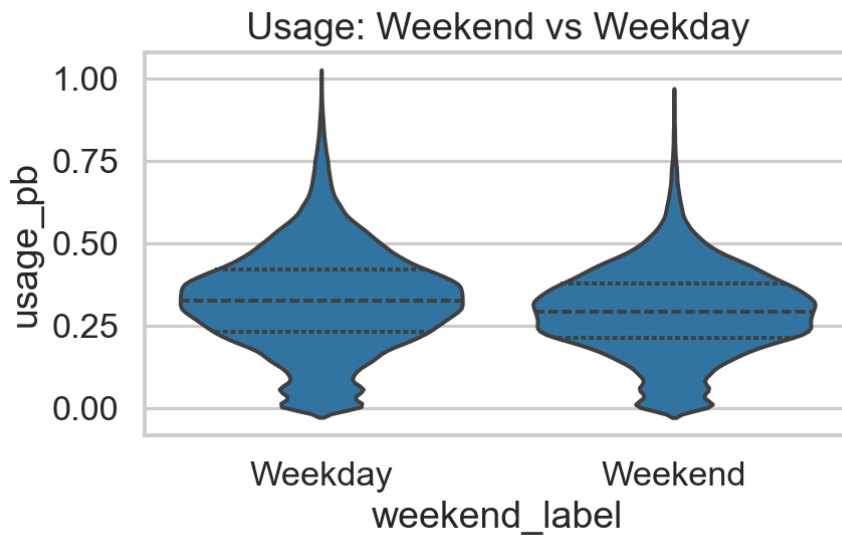


Figure 12: Usage: Weekend vs Weekday.

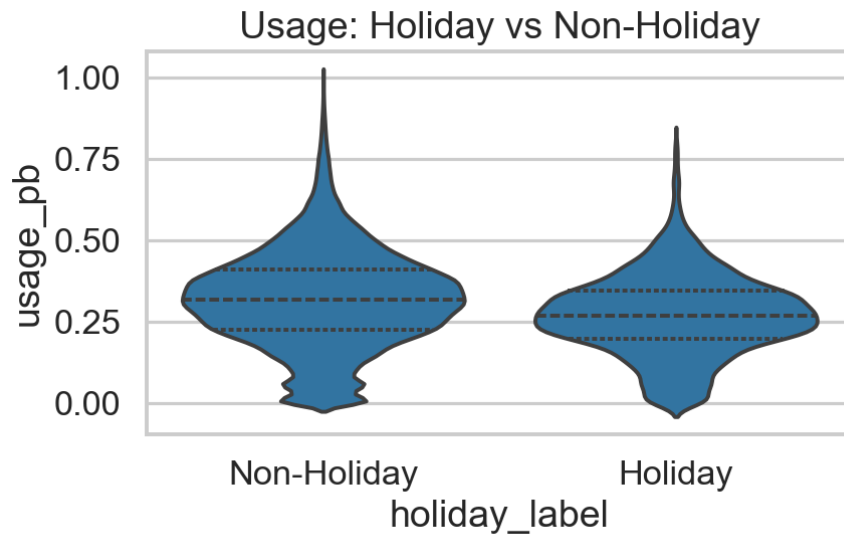


Figure 13: Usage: Holiday vs Non-Holiday.

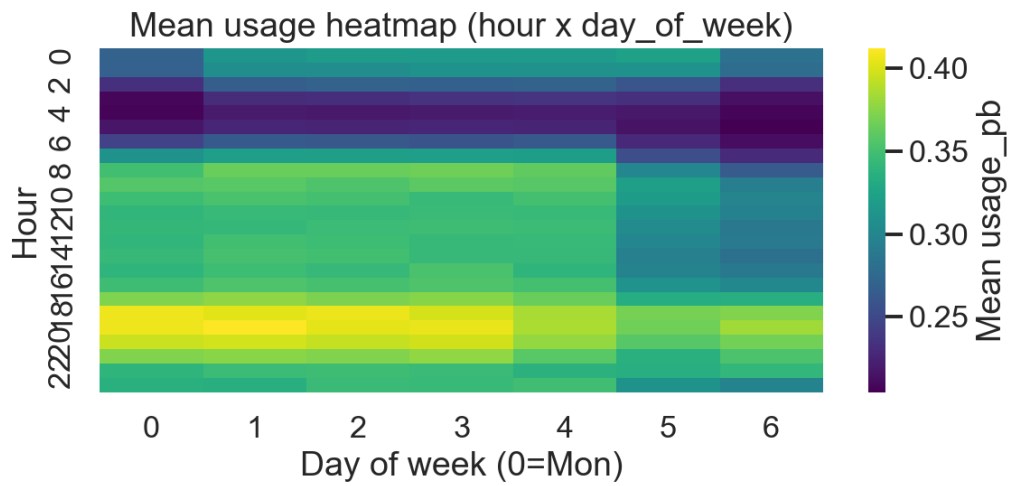


Figure 14: Mean usage heatmap (hour x day_of_week).

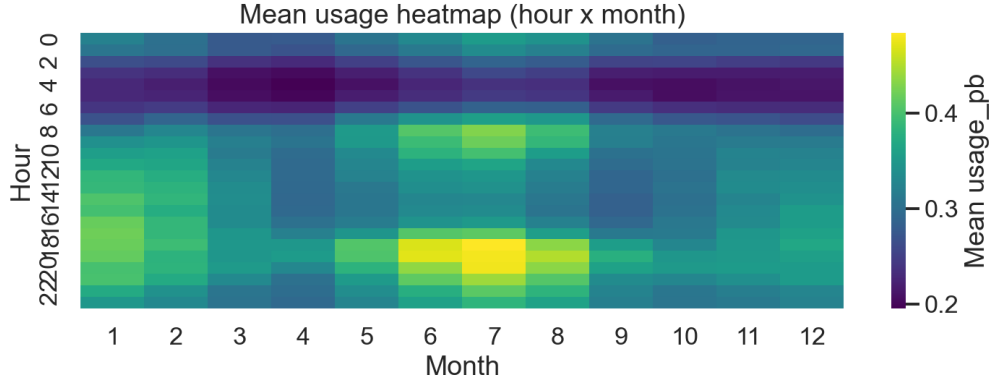


Figure 15: Mean usage heatmap (hour x month).

Takeaway (Demand regimes) The distribution and seasonality plots show clear diurnal structure and moderate seasonal shifts, indicating that hour-of-day and lagged usage features are strong predictors. Weekend and holiday effects are smaller and act as secondary signals.

10.3 Correlation Analysis

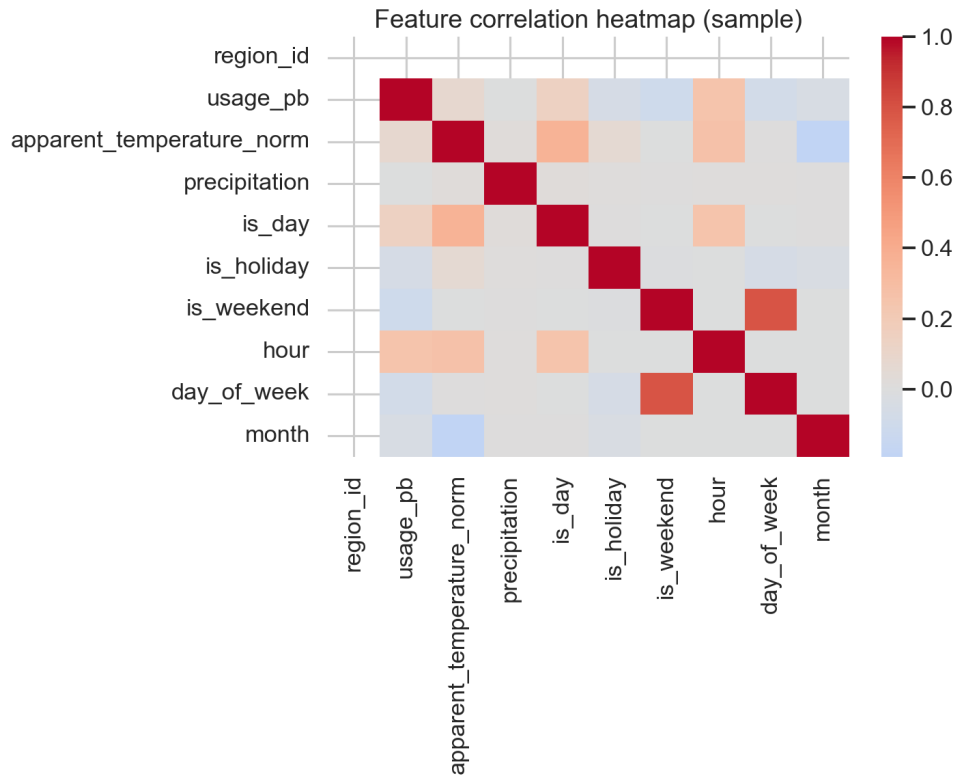


Figure 16: Feature correlation heatmap (sample).

Takeaway (Feature relationships) The correlation heatmap shows limited multicollinearity among engineered features, reducing the risk that importance estimates are driven by redundant signals.

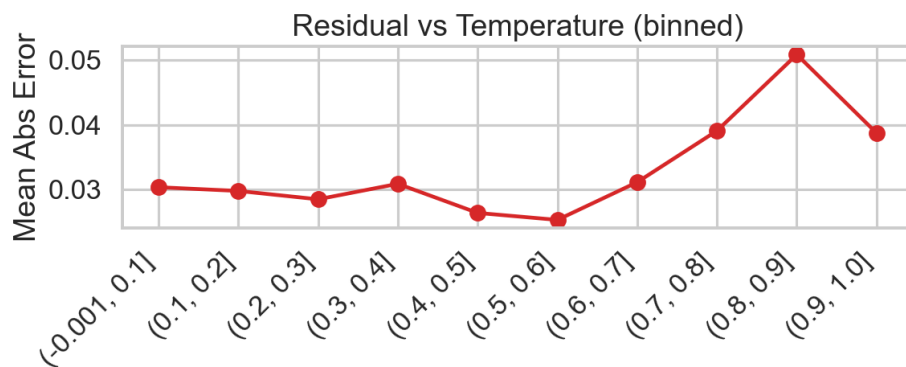


Figure 17: Residual vs Temperature (binned MAE).

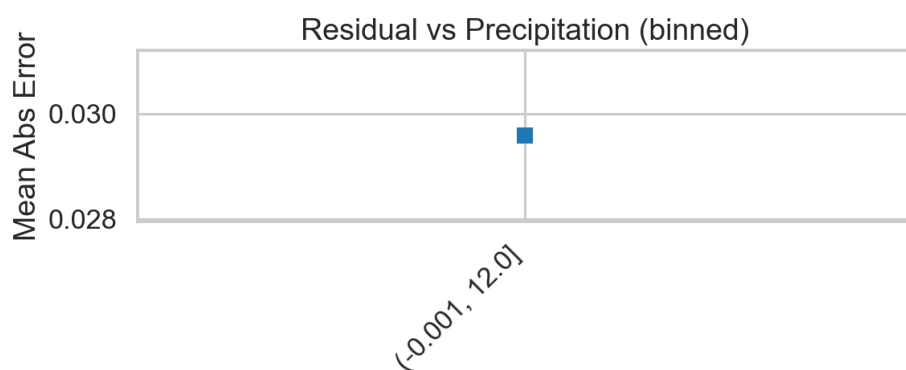


Figure 18: Residual vs Precipitation (binned MAE).

Takeaway (Where the model fails) Errors increase in specific temperature bins and during transition hours, indicating that a single global model may underfit rare, spike-like regimes. This motivates richer exogenous inputs and segment-aware modeling for high-risk periods.

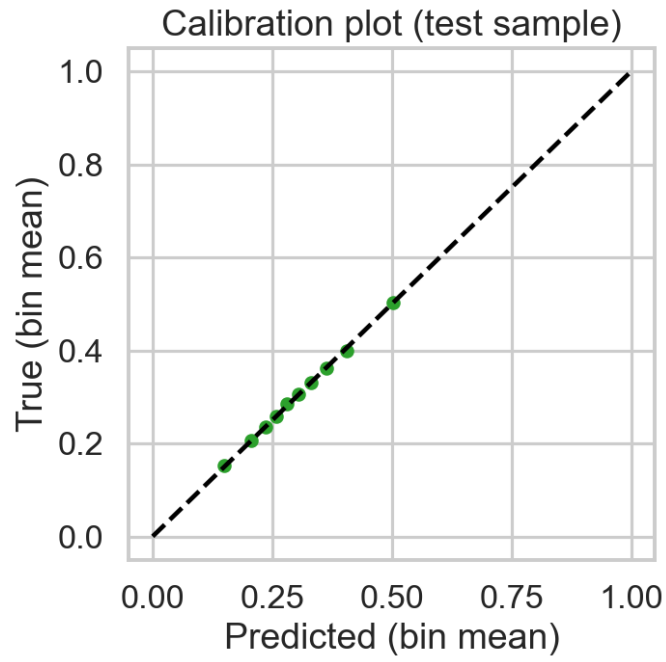


Figure 19: Calibration: predicted deciles vs true mean.

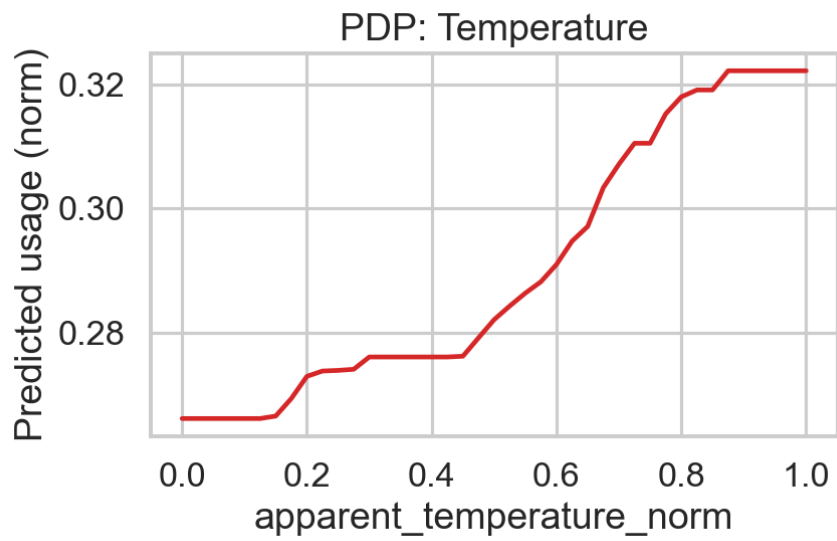


Figure 20: PDP: Temperature effect on predicted usage.

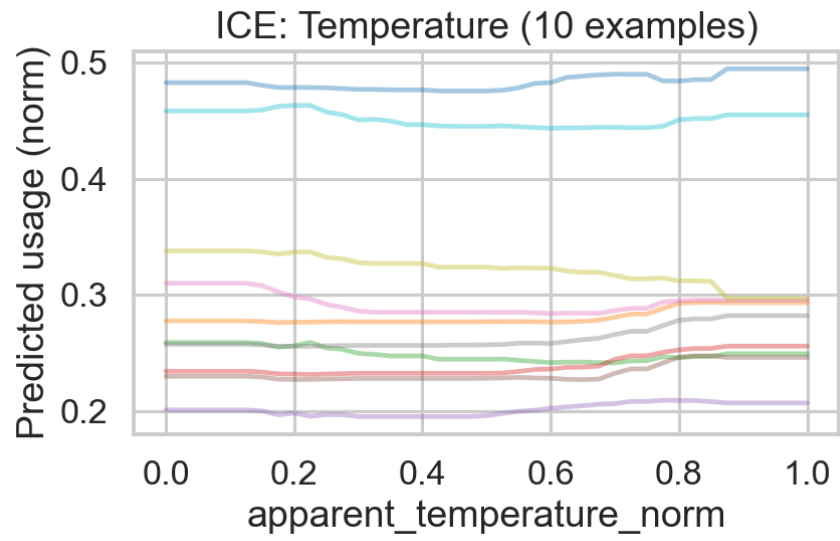


Figure 21: ICE: Temperature (10 examples).

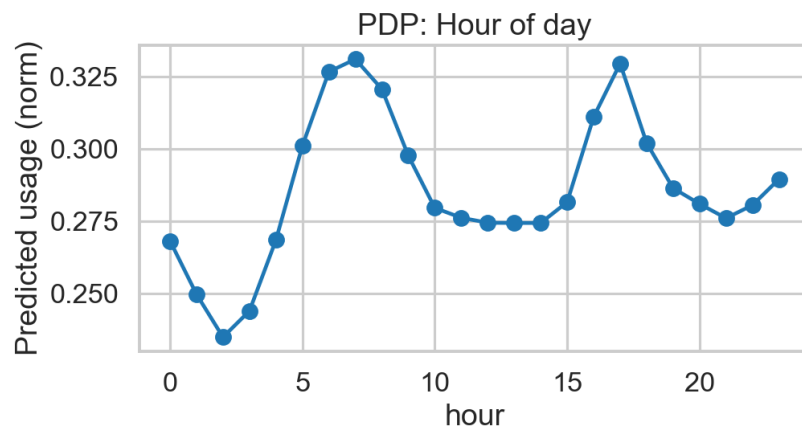


Figure 22: PDP: Hour of day.

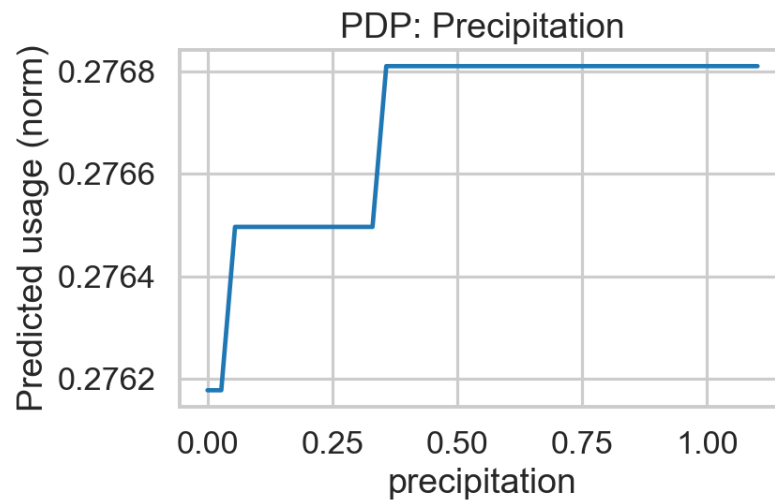


Figure 23: PDP: Precipitation.

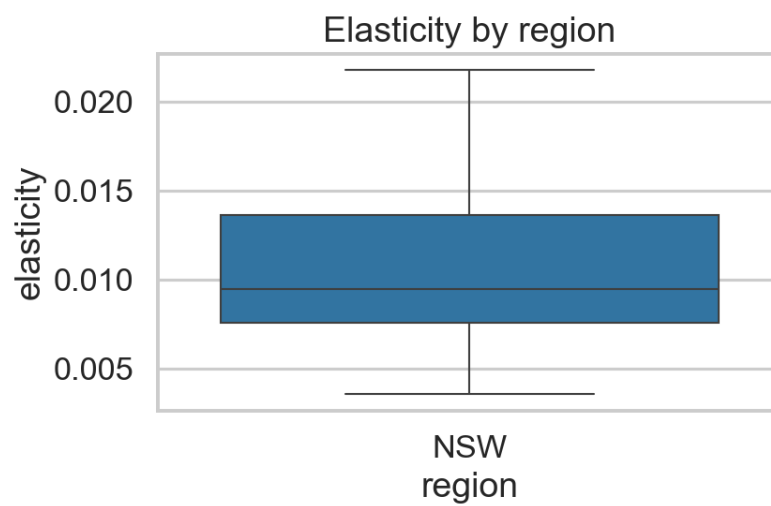


Figure 24: Elasticity by region (boxplot).

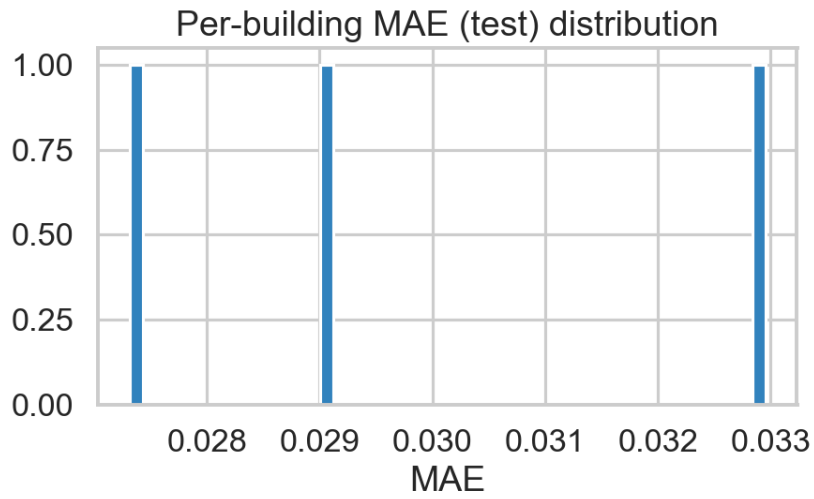


Figure 25: Per-building MAE distribution (test).

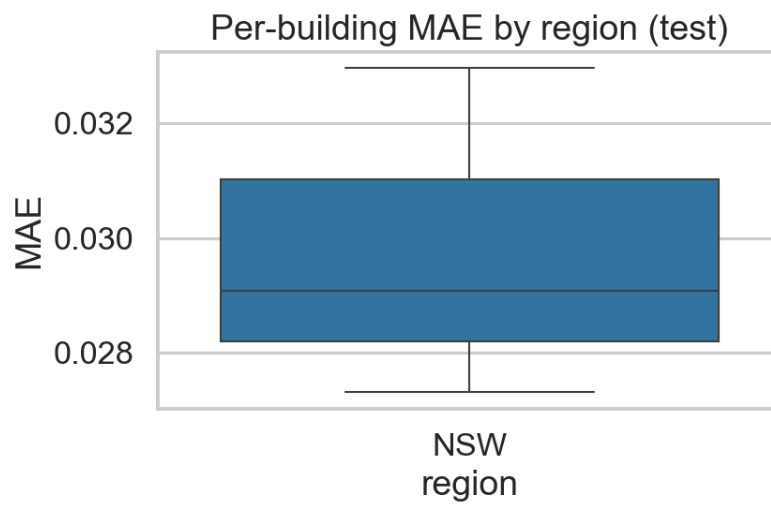


Figure 26: Per-building MAE by region (test).

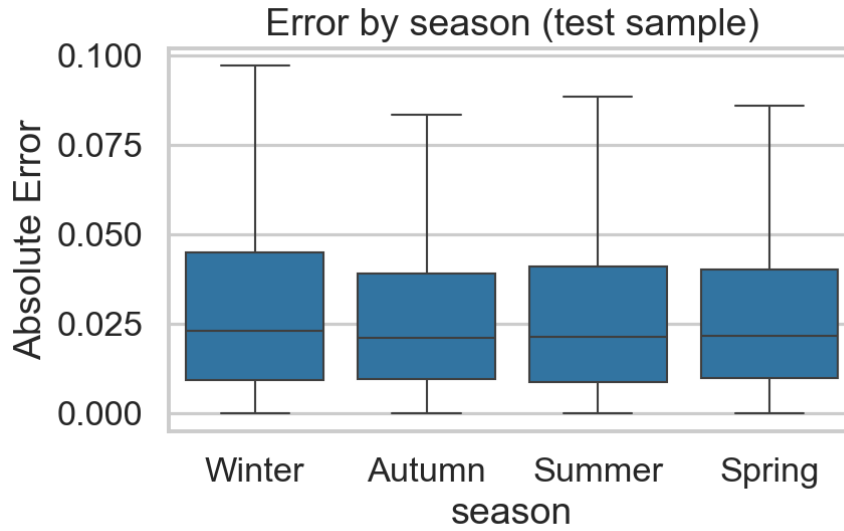


Figure 27: Error by season (test sample).

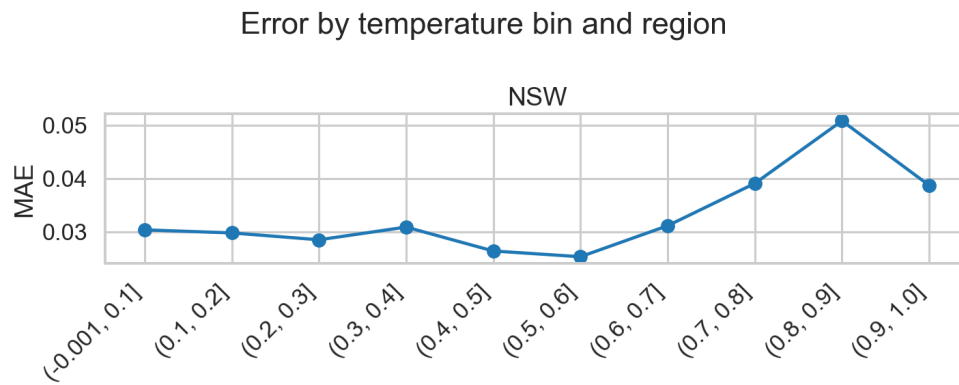


Figure 28: Error by temperature bin and region. Temperature regimes differ by region; higher bins align with larger errors in hotter periods.

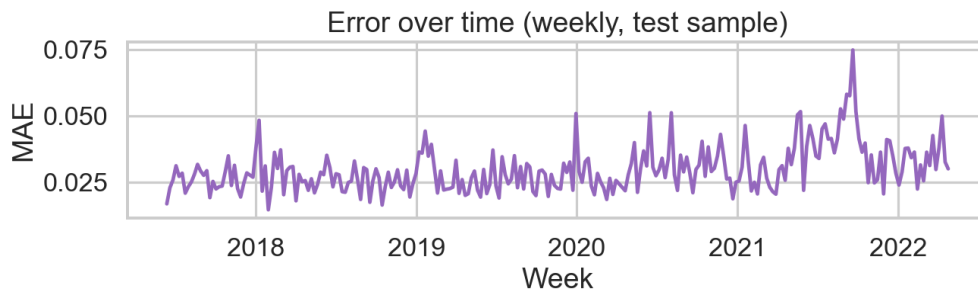


Figure 29: Error over time (weekly MAE, test sample). Stable weekly MAE with mild variation suggests good temporal generalization.

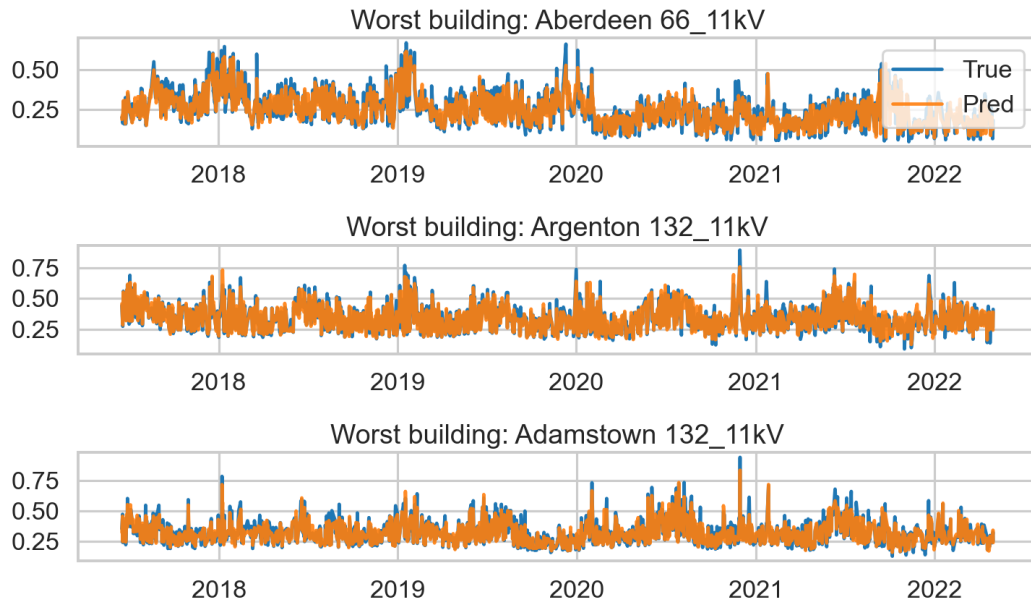


Figure 30: Worst buildings (by MAE): predicted vs actual time series.

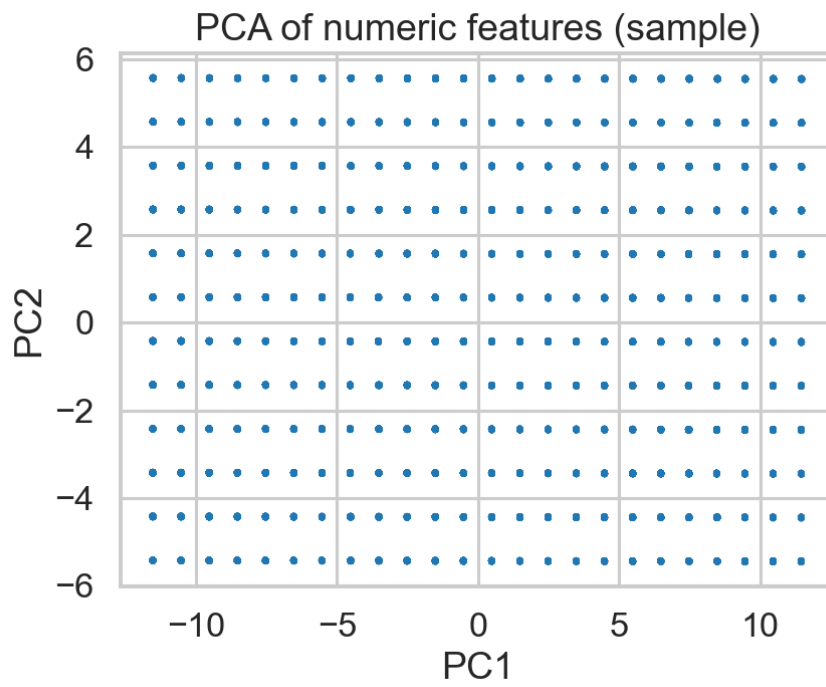


Figure 31: PCA of numeric features (sample).

Why PCA and clustering are included PCA and simple clustering are included as sanity checks for heterogeneous usage profiles. If distinct groups emerge, segment-aware models or elasticity-based stratification may be justified.

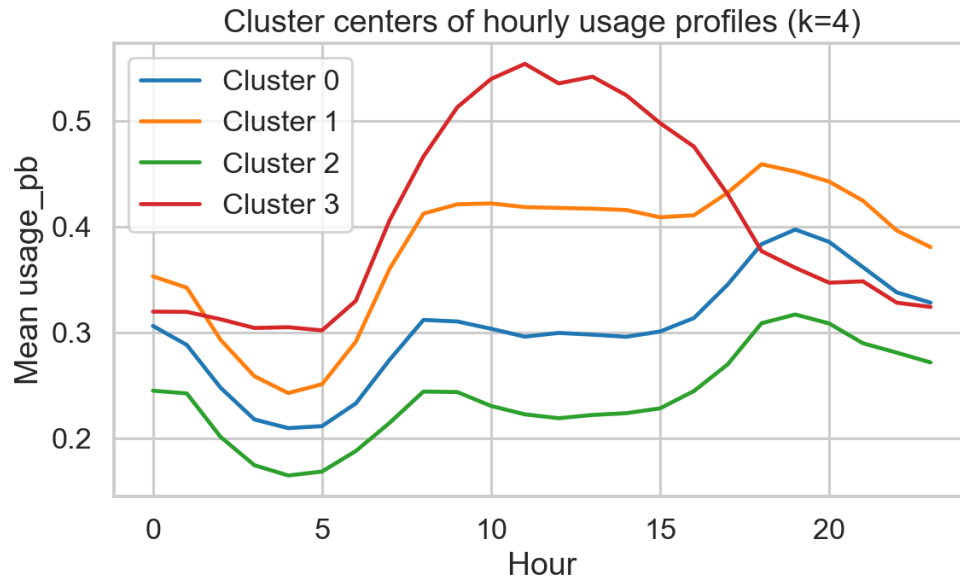


Figure 32: Cluster centers of hourly usage profiles (k=4).

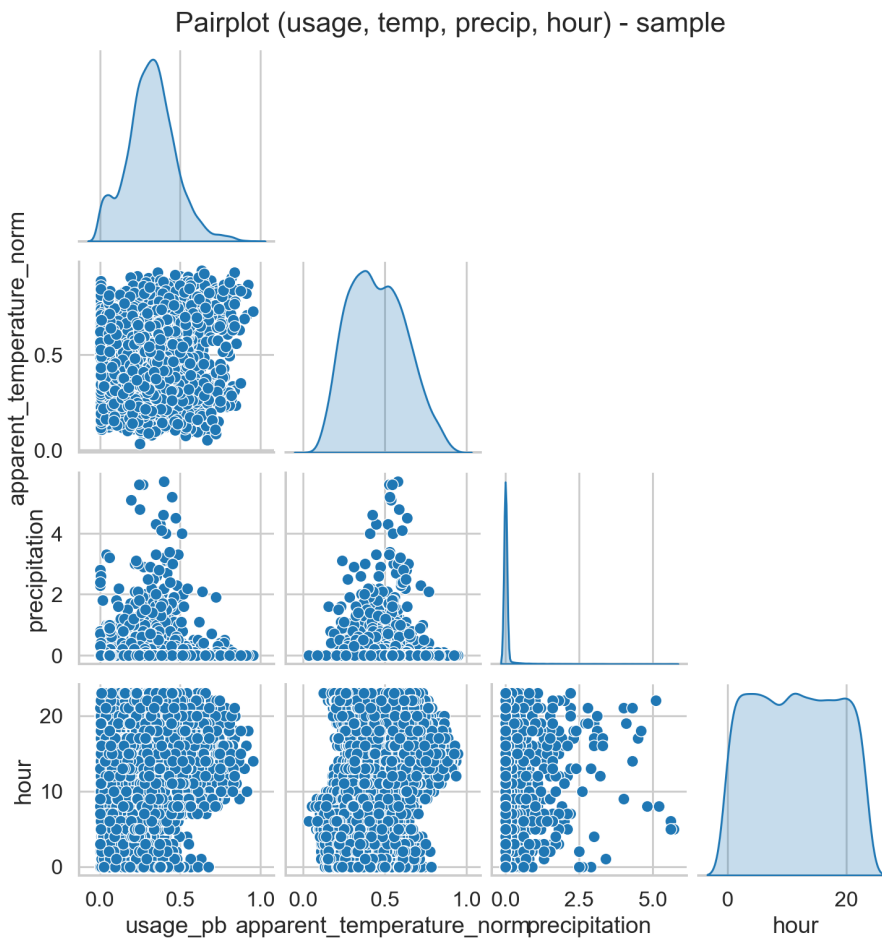


Figure 33: Pairplot (usage, temp, precip, hour) on a sample.

10.4 Model Selection and Training

Model comparisons are reported in the Results section; here we focus on data diagnostics.

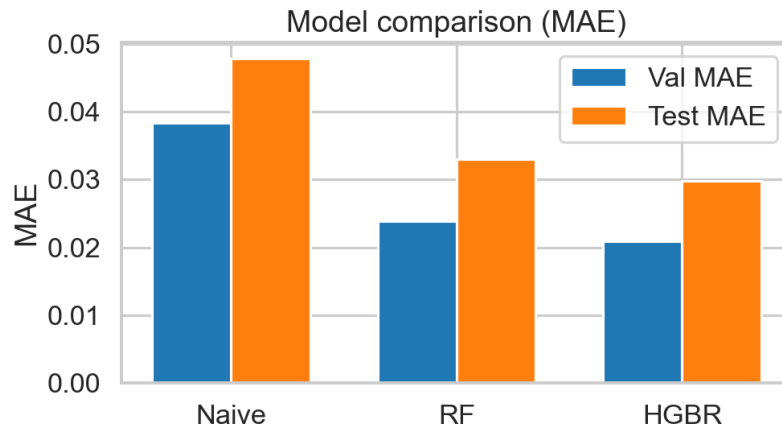


Figure 34: Model comparison (MAE) on validation vs test. HistGBR chosen as primary model.

EDA implication The diagnostics indicate heterogeneous demand regimes, which is reflected in building-level elasticity differences.

11 Model Configuration

11.1 RandomForestRegressor

- `n_estimators=200`, `max_depth=12`, `min_samples_leaf=5`
- `max_features="sqrt"`, `n_jobs=-1`, `random_state=42`

11.2 HistGradientBoostingRegressor (primary)

- `loss="squared_error"`, `learning_rate=0.05`, `max_depth=8`
- `max_iter=300`, `l2_regularization=0.0`, `random_state=42`

12 Model Testing and Deployment

12.1 Prediction Script

12.1.1 Purpose and Prerequisites

The prediction CLI produces next-hour predictions per building from a usage file (CSV/Parquet) and optional weather file. It expects hourly timestamps, building identifiers, and usage; with optional apparent temperature and precipitation for higher fidelity.

12.1.2 Workflow

1) Load the trained model and minimal preprocessing; 2) align/merge weather if provided; 3) compute leakage-safe lags per building for the last hour; 4) predict next-hour usage and write a CSV.

12.1.3 Command-Line Usage Example

```
source .venv/bin/activate && python scripts/predict.py \  
  --input data/ausgrid_with_weather_normalized.csv \  
  --region NSW \  
  --out outputs/tables/predictions_next_hour.csv
```

The pipeline is fully scripted and reproducible. Key steps:

- **Unify data:** `scripts/00_unify_datasets.py` produces a single CSV with region labels.
- **Build features:** `scripts/01_build_features.py` creates leakage-safe features and the next-hour target.
- **Train:** `scripts/02_train_models.py` performs chronological splits and trains Naive/RF/HistGBR.
- **Predict:** `scripts/predict.py` predicts next-hour usage for each building given a usage file (+ optional weather).
- **Explain:** `scripts/03_explain_segments.py` computes temperature elasticity and segments.
- **Plots:** `scripts/04_make_plots.py` + `scripts/04a_extra_plots.py` generate >30 figures.
- **Report:** `scripts/05_render_report.py` assembles this LaTeX report.

See the root `README.md` for exact commands and prediction usage examples.

12.2 Deployment Considerations

- **Internal validity:** enforce leakage controls via per-building shifts/rolling; avoid global scalers fit on full data.
- **External validity:** distribution shifts (policy changes, technology adoption) may degrade accuracy.
- **Bias:** uneven station coverage may bias elasticity; monitor subgroup metrics and confidence intervals.
- **Privacy:** anonymized identifiers; avoid re-identification; follow GDPR principles.

13 Conclusion

Implications for modeling Heterogeneous usage profiles indicate that a single global model can obscure building-level response differences, motivating segment-aware forecasting or elasticity-based stratification.

References

- Pedregosa et al., *Scikit-learn: Machine Learning in Python*, JMLR.
- Ke et al., *LightGBM: A Highly Efficient Gradient Boosting Decision Tree* (future work).
- Lundberg & Lee, *A Unified Approach to Interpreting Model Predictions* (future SHAP enhancements).